

A trace formula for Hecke operators on Maass forms of prime level with primitive character

Dula

6 October 2026

Abstract

We state a Selberg trace formula for T_n , $(n, p) = 1$, on Maass cusp forms of prime level p with a primitive even character χ . The published formula with character (Booker–Lee–Strömbergsson, Theorem 1.7) covers only $n = \pm 1$, and Seymour–Howell’s formula for general n assumes trivial character. The new terms are the split hyperbolic classes, the continuous spectrum twisted by the Hecke eigenvalues of the Eisenstein series, and the parabolic classes for square n . Each term is derived, and Section 3 produces each normalising constant by a computation, given four standard inputs (S1)–(S4). The constants were first read off fits at level 11, and the computation reproduces them exactly. Level 7, with a character of order 3, was a prediction with nothing refitted, and it agrees to 10^{-8} . (S2)–(S4) are proved in appendices. The status is [N] until a human referee has checked the derivation. We then use it for a floating-point pilot of the uniqueness count at level 1951. The pilot says that the certified form near $\frac{1}{4}$ is the only newform in a window around it. Its ratio is 0.78, where any value below 1 suffices. It does not bear on Artin’s conjecture for ρ directly, and Artin’s conjecture for ρ remains open.

1 Setting

p is prime and χ is a primitive character mod p with $\chi(-1) = 1$. A Maass form satisfies $f(\gamma z) = \chi(d)f(z)$ for $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_0(p)$. The Hecke operator $T_n f(z) = n^{-1/2} \sum_{ad=n} \chi(a) \sum_{b \bmod d} f\left(\frac{az+b}{d}\right)$ has eigenvalues $\lambda_j(n)$ on the newforms u_j , with $\lambda_j = \frac{1}{4} + r_j^2$. All forms are new, since χ is primitive and p is prime. Let h be even and holomorphic in a strip, and let $g(u) = \frac{1}{2\pi} \int h(r) e^{-iru} dr$. Write $L(1, \psi_D)$ for Zagier’s L -value of a nonsquare discriminant $D = dl^2$: $L(1, \psi_D) = L(1, \psi_d) l^{-1} \prod_{q \mid l} (1 + (q - \psi_d(q)) \frac{q^e - 1}{q - 1})$.

2 The formula

Theorem 1 ([N]). *For $(n, p) = 1$,*

$$\sum_j h(r_j) \lambda_j(n) = \mathcal{H}_n + \mathcal{C}_n + \mathcal{S}_n + \mathcal{Q}_n,$$

where the four terms are as follows.

Non-split classes. Here $D = t^2 - 4n$ runs over nonsquares, and $H_{t,n}(\chi)$ is the local factor of [1, Thm. 1.7] at p :

$$\mathcal{H}_n = \sum_{\substack{t \in \mathbb{Z} \\ D > 0}} H_{t,n} L(1, \psi_D) g\left(2 \log \frac{|t| + \sqrt{D}}{2\sqrt{n}}\right)$$

$$+ \sum_{D < 0} H_{t,n} L(1, \psi_D) \frac{\sqrt{|D|/n}}{\pi} \int_0^\infty \frac{g(u) \cosh(u/2)}{4 \sinh^2(u/2) + |D|/n} du.$$

Continuous spectrum. Let $\lambda(n, r) = \sum_{ad=n} \chi(d)(a/d)^{ir}$, the Hecke eigenvalue of $E_{1,\chi}$. Then

$$\begin{aligned} \mathcal{C}_n &= \frac{1}{2\pi} \int_{\mathbb{R}} h(r) \lambda(n, r) (-\psi(\tfrac{1}{2} + ir) - \psi(\tfrac{1}{2} - ir)) dr \\ &+ 2 \sum_{m \geq 2} \frac{\Lambda(m)}{m} \sum_{ad=n} \chi(d) (\bar{\chi}(m) g(\log \tfrac{a}{d} + 2 \log m) + \chi(m) g(\log \tfrac{a}{d} - 2 \log m)). \end{aligned}$$

Split classes. Here $L = \log(a/d)$ and $m = |a - d|$, and the sum runs over ordered pairs:

$$\begin{aligned} \mathcal{S}_n &= \sum_{\substack{ad=n \\ a \neq d}} \frac{\chi(a) + \chi(d)}{2} \left[x_p(m) g(L) + \tfrac{1}{2} J(L) \right], \\ J(L) &= -4 \int_{|L|}^\infty g'(u) \log \frac{1 + \sinh(u/2)/\sinh(|L|/2)}{2} du, \\ x_p(m) &= \frac{2}{m} \sum_{q|m} \varphi(q) \log q + \frac{\log p}{m} \sum_{q|m, p \nmid q} \varphi(q) + 2 \log \pi - 3 \log p. \end{aligned}$$

Square classes. Here $n = k^2$; otherwise $\mathcal{Q}_n = 0$. $R_1(g)$ is the $n = 1$ remainder:

$$\begin{aligned} \mathcal{Q}_{k^2} &= \chi(k) [R_1(g) + 2g(0) \log k] + \frac{\chi(k)}{k} I(g), \quad I(g) = -\frac{p+1}{6} \int_0^\infty \frac{g'(u)}{\sinh(u/2)} du, \\ R_1(g) &= 2 \left[(2\gamma + \log 4\pi + \tfrac{1}{2} \log 2 - \tfrac{3}{2} \log p) g(0) - \int_0^\infty \log(\sinh \tfrac{u}{2} \tanh \tfrac{u}{4}) g'(u) du \right] \\ &\quad - g(0) \log 2 + \frac{1}{\pi} \int_{\mathbb{R}} h(r) \operatorname{Re} \psi(\tfrac{1}{2} + ir) dr. \end{aligned}$$

At $n = 1$ the theorem reduces to [1, Thm. 1.7] in the case $s = e = 1$. One correction is needed there: the $\log 2$ coefficient of $g(0)$ is 1, which our transcription of the theorem garbles.

3 Derivation of the three normalising constants

Conventions as in Iwaniec, *Spectral methods*: $u(z, w) = |z - w|^2 / (4 \operatorname{Im} z \operatorname{Im} w)$, $Q(v) = \int_{\mathbb{R}} k(v + t^2) dt$, $g(u) = 2Q(\sinh^2 \tfrac{u}{2})$, and $h(r) = \int g(u) e^{iru} du$. The kernel of T_n is $n^{-1/2} \sum_{\gamma} \chi(a_{\gamma}) k(z, \gamma w)$, where a_{γ} is the upper-left entry. This matches the definition of T_n , with $\chi(a)$ on $\frac{az+b}{d}$, and $a_{\gamma} \bmod p$ is invariant under $\Gamma_0(p)$ -conjugation. The sum runs over $\gamma \in \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_2(\mathbb{Z}) : ad - bc = n, p \mid c \right\} / \{\pm 1\}$. We integrate over F_Y , the fundamental domain with both cusps cut at height Y , and let $Y \rightarrow \infty$. The proof uses four standard inputs. They are cited here, not re-proved:

(S1) Truncation and the Maass–Selberg relations: Iwaniec, Ch. 6 and 10.

(S2) The scattering matrix for primitive χ : Φ is anti-diagonal, with $\varphi_{0\infty}(s) = \sqrt{\pi} \frac{\Gamma(s - \frac{1}{2})}{\Gamma(s)} p^{-s} \frac{L(2s-1, \chi)}{L(2s, \chi)}$ and $\varphi_{\infty 0}$ the same with $\bar{\chi}$. It is unitary on the line. This is *proved* in Appendix A.

- (S3) The Hecke eigenvalue of E_∞ is $\sum_{ad=n} \chi(a)(a/d)^{s-\frac{1}{2}}$, and E_∞ pairs with $\varphi_{\infty 0}$. Equivalently, $\lambda(n, r) = \sum_{ad=n} \chi(d)(a/d)^{ir}$ pairs with $\omega_A = \frac{d}{ds} \log \varphi_{0\infty}$. This is *proved* in Appendix B.
- (S4) The local factor $H_{t,n}(\chi)$ for nonsplit classes, $(n, p) = 1$. It is the χ -weighted count of $\Gamma_0(p)$ -classes, $\sum_f h_w(D/f^2)\mu(t, f, n)$, normalised by $\sum_f h_w(D/f^2)$. This is *proved* in Appendix C for both signs of D , and checked by brute force in 9478 cases with $D < 0$ and 21590 with $D > 0$.

Continuous spectrum and the constant $2 \log \pi - 3 \log p$. By (S1) the continuous spectrum contributes $\frac{1}{4\pi} \sum_X \int h(r) \lambda_X(n, r) (2 \log Y - \omega_X(r)) dr$ to the truncated trace. Here X runs over the Hecke basis $\{E_A, E_B\}$, with $\lambda_B(n, r) = \lambda_A(n, -r)$ and $\omega_B(r) = \omega_A(-r)$, and $\omega_A(r) = \frac{d}{ds} \log \varphi_{0\infty}(s)$ at $s = \frac{1}{2} + ir$ (Appendix B gives the pairing). Because Φ has zero diagonal, there are no $Y^{\pm 2ir}$ terms and no $h(0)$ term. Since $\int h(r)(a/d)^{ir} dr = 2\pi g(\log \frac{a}{d})$, the $\log Y$ part equals $2 \log Y \sum_{ad=n} \chi(d) g(\log \frac{a}{d})$. By (S2), $\omega_A = \psi(s - \frac{1}{2}) - \psi(s) - \log p + 2 \frac{L'}{L}(2s - 1, \chi) - 2 \frac{L'}{L}(2s, \chi)$. Apply the functional equation of $L(w, \chi)$,

$$\frac{L'}{L}(w, \chi) = -\log \frac{p}{\pi} - \frac{1}{2} \psi\left(\frac{w}{2}\right) - \frac{1}{2} \psi\left(\frac{1-w}{2}\right) - \frac{L'}{L}(1-w, \bar{\chi}),$$

at $w = 2ir$. The terms $\psi(ir)$ cancel, and we get

$$\omega_A(r) = (2 \log \pi - 3 \log p) - \psi\left(\frac{1}{2} + ir\right) - \psi\left(\frac{1}{2} - ir\right) + 2 \sum_{m \geq 2} \frac{\Lambda(m)}{m} (\bar{\chi}(m) m^{2ir} + \chi(m) m^{-2ir}).$$

The constant $2 \log \pi - 3 \log p$ is therefore $-\log p$ (from p^{-s}) plus $-2 \log(p/\pi)$ (from the functional equation). It multiplies $\sum_{ad=n} \chi(d) g(\log \frac{a}{d})$, and for $a \neq d$ it is collected into $x_p(m)$. (At level 11 this constant was first read off a fit, -4.90422 . The derivation gives it exactly, and level 7 was a prediction.)

Split classes: the orbital integral and the coefficient $\frac{1}{2}$ of J . A split class is conjugate to

$$\gamma = \begin{pmatrix} \alpha & \beta \\ 0 & \delta \end{pmatrix} \text{ with } \alpha\delta = n, \alpha \neq \delta, \text{ and } \beta \text{ taken mod } m = |\alpha - \delta|. \text{ It fixes } \infty \text{ and } \xi = \beta/(\delta - \alpha) = r/q,$$

with $q \mid m$. Its centraliser in Γ is $\{\pm 1\}$, so the class contributes $\int_{\mathbb{H}_Y} k(z, \gamma z) d\mu$. Here $\mathbb{H}_Y$ is $\mathbb{H}$ with all horoballs removed. Only the horoballs at ∞ and at ξ matter as $Y \rightarrow \infty$. The horoball at ξ has diameter $\delta_\xi = (q^2 Y)^{-1}$ if $p \mid q$ (cusp ∞) and $(pq^2 Y)^{-1}$ if $p \nmid q$ (cusp 0, width p). Translate ξ to 0, so that γ becomes $z \mapsto \lambda z$ with $\lambda = \alpha/\delta$. In polar coordinates $z = \rho e^{i\theta}$ the region is $\delta_\xi \sin \theta \leq \rho \leq Y/\sin \theta$, and $u(z, \lambda z) = c/\sin^2 \theta$ with $c = \sinh^2 \frac{L}{2}$, $L = \log \lambda$. Below, $\sinh \frac{L}{2}$ means $\sinh \frac{|L|}{2}$, since ordered pairs with $\alpha < \delta$ have $L < 0$. Hence

$$\int k(z, \gamma z) d\mu = \frac{Q(c)}{\sqrt{c}} \log \frac{Y}{\delta_\xi} + J_k, \quad J_k = \int_{\mathbb{R}} k(c(1+v^2)) \log(1+v^2) dv.$$

Abel inversion $k(x) = -\frac{1}{\pi} \int_x^\infty Q'(t)(t-x)^{-1/2} dt$, together with $\int_c^t \log \frac{x}{c} \frac{dx}{\sqrt{(x-c)(t-x)}} = 2\pi \log \frac{1+\sqrt{t/c}}{2}$, gives

$$J_k = -\frac{2}{\sqrt{c}} \int_c^\infty Q'(t) \log \frac{1+\sqrt{t/c}}{2} dt = \frac{J(L)}{4 \sinh(L/2)}, \quad Q(c)/\sqrt{c} = \frac{g(L)}{2 \sinh(L/2)}.$$

Now sum over $\beta \bmod m$, using $\#\{\beta : q_\beta = q\} = \varphi(q)$, $\log(Y/\delta_\xi) = 2 \log Y + 2 \log q + [p \nmid q] \log p$, and $m/(2 \sinh \frac{L}{2}) = \sqrt{n}$:

$$\sum_\beta \int k(z, \gamma_\beta z) d\mu = \sqrt{n} \left[2g(L) \log Y + g(L)(B(m) + f(m) \log p) + \frac{1}{2} J(L) \right],$$

where $B(m) = \frac{2}{m} \sum_{q|m} \varphi(q) \log q$ and $f(m) = \frac{1}{m} \sum_{q|m, p \nmid q} \varphi(q)$. Classes are enumerated once per fixed point. A fixed point Γ -equivalent to ∞ gives an upper-triangular representative, as above.

A fixed point equivalent to 0 is moved to ∞ by the Fricke involution W . Here $W \begin{pmatrix} a & b \\ c & d \end{pmatrix} W^{-1} = \begin{pmatrix} d & -c/p \\ -pb & a \end{pmatrix}$, so W swaps α and δ in the character. W also preserves the horoball system, so the orbital integral is unchanged. Each class has two fixed points and is therefore counted twice. When both fixed points are ∞ -type, $p \mid q \mid m$ forces $\alpha \equiv \delta \pmod{p}$, so the two weights agree. After the factor $n^{-1/2}$, each ordered pair (α, δ) contributes $\frac{\chi(\alpha) + \chi(\delta)}{2} [2g(L) \log Y + g(L)(B + f \log p) + \frac{1}{2} J(L)]$. *Normalisation check:* summed over ordered pairs, the $\log Y$ terms give $2 \log Y \sum_{a \neq d} \chi(d) g(\log \frac{a}{d})$. This is exactly the continuous-spectrum term above, so Y drops out with no free factor, and the $\frac{1}{2}$ on J comes from the same integral as the $\log Y$ term. Adding the constant from the continuous spectrum gives $x_p(m) = B(m) + f(m) \log p + 2 \log \pi - 3 \log p$.

Square classes and the coefficient 2 of $g(0) \log k$. For $n = k^2$ the parabolic classes are $\begin{pmatrix} k & \beta \\ 0 & k \end{pmatrix}$,

$\beta \neq 0$, acting by $z \mapsto z + \beta/k$. Put $P_1(Y) = \sum_{\beta \neq 0} \int_0^Y \int_0^1 k(z, z + \beta) d\mu$. Here P_1 is taken on the strip without excising the other horoballs. This is harmless for compactly supported k , since the integrand vanishes for y below a fixed bound and those horoballs lie below $1/Y$. The substitution $y' = ky$ gives $\sum_{\beta \neq 0} \int_{y \leq Y} k(z, z + \beta/k) d\mu = k P_1(kY)$. After the factor $n^{-1/2} = k^{-1}$, and the same at the cusp 0, the block is $\chi(k) [P(Y) + A \log k]$, where $P(Y) = A \log Y + B_0 + o(1)$ is the $n = 1$ parabolic block. At $n = 1$ the continuous spectrum contributes $2g(0) \log Y$ (pair $(1, 1)$), and Y must drop out, so $A = 2g(0)$. Hence the extra term is $2g(0) \log k$. The identity class kI contributes $\chi(k)k^{-1}$ times the $n = 1$ identity term, since kI acts trivially and $k(z, z) = k(0)$. Finally R_1 is the $n = 1$ block of [1, Thm. 1.7], with the $\log 2$ correction, minus the ψ -part of $\mathcal{C}_1$.

Independent checks of the ingredients.

- The orbital-integral formula, computed by direct two-dimensional quadrature over the truncated region, agrees to 10^{-15} for $k(u) = e^{-3u}$ at three values of (λ, Y, δ) . The two expressions for J_k agree.
- The integral identity holds to 10^{-14} at three (c, t) .
- The functional-equation identity holds to 10^{-16} for χ of order 5 mod 11.

The script is `derive/check_ingredients.py`. An independent review of this section re-derived steps 1–8. It confirmed $2 \log \pi - 3 \log p$ (including unitarity, $|\varphi_{\infty 0}(\frac{1}{2} + ir)| = 1$, which requires p^{-s}), the $\frac{1}{2}$ on J , and $2g(0) \log k$ to 14–30 digits. It found no mathematical error, and its presentation points are applied above.

Status. Given (S1)–(S4), each of the three normalising constants is now produced by a computation, not by a fit. R_1 and its $\log 2$ correction are taken from [1, Thm. 1.7], not derived here. (S1) is textbook. (S2) and (S3) are proved in Appendices A and B, given Iwaniec’s Fourier expansion. (S4) is proved in Appendix C. Theorem 1 nevertheless stays [N] until the derivation, including the imported R_1 , has been checked by a human referee. An independent automated review is not a substitute for that check.

4 Numerical confirmation

Spectra were computed independently with a Hejhal solver (both parities, Hecke relations to 10^{-7}). The two levels are $p = 11$ with χ of order 5 (63 forms, $R \leq 10.5$) and $p = 7$ with χ of order 3 (26 forms, $R \leq 9$). The test functions are Gaussians $h(r) = e^{-r^2/2\sigma^2}$.

what	how it was fixed	check
coefficient $\frac{1}{2}$ of J	fit, $n = 2, 3, 5, 7$ at level 11	0.49998, 0.49995, 0.5004, 0.4998
constant $2 \log \pi - 3 \log p$	fit, same data	-4.90422 against -4.904214
coefficient 2 of $g(0) \log k$	fit at $n = 4$	2.0000000
$n = 6, 8, 10, 12, 14, 15$ (level 11)	prediction	$ \text{spec} - \text{geom} \leq 5 \times 10^{-7}$ (mostly $\sim 10^{-8}$)
$n = 9$ (level 11)	prediction	$\log k$ coefficient 2.0000000
level 7, order-3 χ : $n = 1, \dots, 6, 8, 9$	prediction, nothing refitted	$ \text{spec} - \text{geom} \leq 7 \times 10^{-8}$ ($\leq 1.1 \times 10^{-8}$ at $\sigma = 0.8$)
$n = 8$ at level 7 ($m = 7 = p$)	tests the $p \mid q$ branch of x_p	5.9×10^{-9}
production code, bump g , level 11, $n = 1$	against the completed spectrum (178 eigenvalues, $R \leq 16$)	residual 5.0, 1.5, 0.64×10^{-6} for $a = 6, 8, 10$, i.e. 1.04, 1.04, 1.07 times the Weyl mass above $R = 16$. The earlier excess came from 26 eigenvalues missing from the first list (one cut by the two-height test, 24 missed by a step-0.01 scan or rejected as mixed pairs, and a distinct eigenvalue at $R = 8.697433$, 5.2×10^{-4} from 8.697956, with $t_2 = -0.832$ against +1.464)
Gaussian $\sigma = 3$, level 11, $n = 1$	against the 178 eigenvalues	remainder 4.8×10^{-4} , against a tail of 1.2×10^{-5} above $R = 16$. <i>Open</i> : pairwise single-eigenvalue fits disagree ($R \approx 10.0$ and 12.4), so the missing weight is spread above $R \approx 10$. Not used anywhere in the derivation or the pilot

5 Uniqueness of the form near $\frac{1}{4}$ at level 1951: a pilot

Take $p = 1951$, and let χ be the quintic character of the project, $\chi(n) = e^{4\pi i \text{ind}_3(n)/5}$. For every newform the numbers $t_p = \bar{\chi}(p)^3 \lambda(p)$ are real, since $\chi^5 = 1$, and the Hecke relations give $\lambda(2^e 3^f) = \chi(2^e 3^f)^3 U_e(t_2/2) U_f(t_3/2)$, with U_k the Chebyshev polynomials of the second kind. Let $f(u) = \cos^2(\pi u/a)$ for $|u| \leq a/2$, $g = f * f$ and $h = \hat{f}^2$. Then $h \geq 0$ on $\mathbb{R}$, and $h(it) = \int g \cosh(tu) du \geq 0$, so every spectral weight is non-negative, exceptional eigenvalues included.

Proposition 2 ([T], with u_n defined spectrally). *Let $c = (c_{ab})_{a \leq A, b \leq B}$ be real. Put $u_n = \bar{\chi}(n)^3 \sum_j h(r_j) \lambda_j(n)$, the spectral sum itself, and $Q(c) = \sum_{a,b,a',b'} c_{ab} c_{a'b'} \sum_{e,f} u_{2^e 3^f}$. Here e runs from $|a - a'|$ to $a + a'$ and f from $|b - b'|$ to $b + b'$, both in steps of 2. Let W be the set of (λ, t_2, t_3) with $|r| \leq r_W$ (real or imaginary), $|t_2 - 1| \leq \varepsilon$ and $|t_3 + \varphi| \leq \varepsilon$. If*

$$Q(c) < 2 h_{\min} \min_W \left(\sum c_{ab} U_a\left(\frac{t_2}{2}\right) U_b\left(\frac{t_3}{2}\right) \right)^2, \quad h_{\min} = \min_{|r| \leq r_W} h(r),$$

then at most one newform of level 1951 and character χ lies in W .

Proof. By the Clebsch–Gordan rule $U_a U_{a'} = \sum_e U_e$, we have $Q(c) = \sum_j h(r_j) P_j$ with $P_j = (\sum c_{ab} U_a(t_{2,j}/2) U_b(t_{3,j}/2))^2 \geq 0$. Two distinct newforms in W would each contribute at least $h_{\min} \min_W P$. $\square$

The proposition uses only positivity, so it holds as stated. Applying it requires the spectral sums u_n , and the pilot computes those from Theorem 1, which is [N], using class numbers that assume GRH. *The conclusion of the pilot is therefore [N], not [T].*

The pilot [N]. We take $a = 18$, $A = B = 2$, and the 25 traces with $n \mid 2^4 3^4$. Theorem 1 is evaluated in floating point, using 1.8×10^6 values $L(1, \psi_D)$ with $|D| \leq 2.1 \times 10^{10}$ (PARI, 70 minutes on one core).

- $\max_n |\operatorname{Im} u_n| = 4.5 \times 10^{-10}$, consistent with the reality of t_2, t_3 .
- The optimal $c = M^{-1}v$, where v is the vector of Chebyshev values at $(t_2, t_3) = (1, -\varphi)$, gives $Q = 126.36$, against the threshold $2h_{\min}(c \cdot v)^2 = 162.00$ for the certified window $(|\lambda - \frac{1}{4}| \leq 8.7 \times 10^{-10}, \varepsilon = 2 \times 10^{-13})$. The ratio is 0.780. Degrees (1, 2) alone give 0.793, and degrees (1, 1) give 0.934.
- The same c still gives a ratio below 1 for the much larger window $|\lambda - \frac{1}{4}| \leq 2.5 \times 10^{-3}$, $|t_2 - 1| \leq 0.05$, $|t_3 + \varphi| \leq 0.05$.
- Sanity: $h(0) v^T M^{-1}v = 0.641 \in (\frac{1}{2}, 1]$, and M is positive definite ($\lambda_{\min} = 250$ for $A = B = 2$; 396 for degrees (1, 2)).

Existence of a form in the window is the eigenvalue certificate of the companion paper ([T|A]). Combined with the pilot, this says, numerically, that ϕ is the only newform in the window. It does not say that $\lambda_\phi = \frac{1}{4}$, and it does not say that $\lambda_\phi(p) = a_p(\rho)$ at every prime.

What is missing for [T].

1. A human referee’s check of the whole derivation. (S2)–(S4) are proved in Appendices A–C, and Section 3 computes the three normalising constants from them. The referee should also check two imported items: the $n = 1$ block R_1 with its log 2 correction, taken from [1, Thm. 1.7], and Iwaniec’s Fourier expansion.
2. Unconditional values of $L(1, \psi_D)$. PARI’s `quadclassunit` assumes GRH, and `qfbclassno` is not guaranteed. The 1.8×10^6 values would have to be recomputed by counting reduced forms. That costs $O(|D|^{1/2})$ each, which is feasible in C.
3. Interval arithmetic for the quadratures and the ψ -integral. The margin is 22%, so modest precision suffices.

A The scattering matrix for primitive χ (input (S2))

Convention, the same as in Section 1 and as used by the numerics: $f(\delta z) = \chi(d_\delta) f(z)$ for $\delta \in \Gamma_0(p)$, and T_n carries $\chi(a)$. The cusps are ∞ and 0, with $\sigma_\infty = I$ and $\sigma_0 = \begin{pmatrix} 0 & -1/\sqrt{p} \\ \sqrt{p} & 0 \end{pmatrix}$. Both

are singular for χ : the stabilisers are generated by $\pm \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $\pm \begin{pmatrix} 1 & 0 \\ -p & 1 \end{pmatrix}$. Put $E_a(z, s) = \sum_{\gamma \in \Gamma_a \backslash \Gamma} \bar{\chi}(d_\gamma) \operatorname{Im}(\sigma_a^{-1} \gamma z)^s$. Since $d_{\gamma\delta} \equiv d_\gamma d_\delta \pmod{p}$, we get $E_a(\delta z, s) = \chi(d_\delta) E_a(z, s)$.

Proposition 3 ([T], given Iwaniec's Fourier expansion). $\varphi_{\infty\infty} = \varphi_{00} = 0$ and

$$\varphi_{\infty 0}(s) = \sqrt{\pi} \frac{\Gamma(s - \frac{1}{2})}{\Gamma(s)} p^{-s} \frac{L(2s - 1, \bar{\chi})}{L(2s, \bar{\chi})}, \quad \varphi_{0\infty}(s) = \sqrt{\pi} \frac{\Gamma(s - \frac{1}{2})}{\Gamma(s)} p^{-s} \frac{L(2s - 1, \chi)}{L(2s, \chi)}.$$

Moreover $\varphi_{\infty 0}(s) \varphi_{0\infty}(1 - s) = 1$ and $\varphi_{0\infty}(s) = \overline{\varphi_{\infty 0}(\bar{s})}$. Hence $|\varphi_{\infty 0}(\frac{1}{2} + ir)| = |\varphi_{0\infty}(\frac{1}{2} + ir)| = 1$. With p^{-2s} in place of p^{-s} the first product would be p^{-1} .

Proof. By Iwaniec, §3.4, $\varphi_{ab}(s) = \sqrt{\pi} \frac{\Gamma(s - \frac{1}{2})}{\Gamma(s)} \sum_{C > 0} C^{-2s} S_{ab}(0, 0; C)$. Here $S_{ab}(0, 0; C) = \sum \bar{\chi}(d_\gamma)$, summed over the double cosets $\Gamma_\infty \backslash \sigma_a^{-1} \Gamma \sigma_b / \Gamma_\infty$ with lower-left entry C .

$(\infty, 0)$. For $\gamma = \begin{pmatrix} a & b \\ pc' & d \end{pmatrix}$, $\gamma\sigma_0 = \begin{pmatrix} b\sqrt{p} & -a/\sqrt{p} \\ d\sqrt{p} & -c'\sqrt{p} \end{pmatrix}$. So $C = d\sqrt{p}$ with $d \geq 1$ and $(d, p) = 1$.

Right translation by Γ_∞ is $\gamma \mapsto \gamma \begin{pmatrix} 1 & 0 \\ -pn & 1 \end{pmatrix}$, which fixes d and sends $c' \mapsto c' - nd$. The left action fixes the bottom row. The double cosets with this C are indexed by $c' \bmod d$ with $(c', d) = 1$, there are $\varphi(d)$ of them, and each has weight $\bar{\chi}(d)$. This gives $p^{-s} \sum_{(d,p)=1} \varphi(d) \bar{\chi}(d) d^{-2s} = p^{-s} L(2s - 1, \bar{\chi}) / L(2s, \bar{\chi})$.

$(0, \infty)$. $\sigma_0^{-1} \gamma = \begin{pmatrix} c'\sqrt{p} & d/\sqrt{p} \\ -a\sqrt{p} & -b\sqrt{p} \end{pmatrix}$. So $C = |a|\sqrt{p}$, the classes are indexed by $b \bmod a$, and the weight is $\bar{\chi}(d) = \chi(a)$. This gives $p^{-s} L(2s - 1, \chi) / L(2s, \chi)$.

Diagonal entries. For (∞, ∞) we have $p \mid C$, and the weights $\bar{\chi}(d)$ over $d \bmod C$, $(d, C) = 1$, sum to 0. The same holds for $(0, 0)$.

Unitarity. With $\Lambda(w, \psi) = (p/\pi)^{w/2} \Gamma(\frac{w}{2}) L(w, \psi) = \epsilon_\psi \Lambda(1 - w, \bar{\psi})$ and $\epsilon_\chi \epsilon_{\bar{\chi}} = \chi(-1) = 1$,

$$\frac{L(2s - 1, \bar{\chi}) L(1 - 2s, \chi)}{L(2s, \bar{\chi}) L(2 - 2s, \chi)} = \frac{p}{\pi} \cdot \frac{\Gamma(s) \Gamma(1 - s)}{\Gamma(s - \frac{1}{2}) \Gamma(\frac{1}{2} - s)}.$$

Multiplying by the two Gamma prefactors and by $p^{-s} p^{-(1-s)}$ gives $(p/\pi) \cdot \pi \cdot p^{-1} = 1$; with p^{-2s} it would give p^{-1} . Conjugating coefficients gives the second relation. $\square$

Check (derive/check_S2.py; $p = 11$, χ of order 5; $p = 7$, χ of order 3): the product is 1 to 10^{-16} at generic complex s , against $1/p$ for p^{-2s} .

B The Eisenstein eigenvalue and its pairing (input (S3))

Proposition 4 ([T], given Iwaniec's Fourier expansion and (S1)). *Let $(n, p) = 1$. Then E_∞ is an eigenfunction of T_n with eigenvalue*

$$\lambda_\infty(n, s) = \sum_{ad=n} \chi(a) (a/d)^{s-\frac{1}{2}}.$$

In the truncated trace it pairs with $\omega_\infty = \frac{d}{ds} \log \varphi_{\infty 0}$. Consequently the continuous-spectrum term of Section 3 is $\frac{1}{2\pi} \int h(r) \lambda(n, r) \omega_A(r) dr$, with $\lambda(n, r) = \sum_{ad=n} \chi(d) (a/d)^{ir}$ and $\omega_A = \frac{d}{ds} \log \varphi_{0\infty}$ at $s = \frac{1}{2} + ir$.

Proof. Eigenvalue. By Iwaniec, Thm. 3.4, the n -th coefficient of E_∞ is $\pi^s \Gamma(s)^{-1} |n|^{s-1} D(n, s)$, where $D(n, s) = \sum_{p|c} c^{-2s} S(c)$ and $S(c) = \sum_{d \bmod c, (d, c)=1} \bar{\chi}(d) e(nd/c)$. Write $c = p^k m$ with $(m, p) = 1$. By the Chinese remainder theorem, $S(c) = G_k \cdot c_m(n)$, where c_m is Ramanujan's sum and $G_k = \sum_{d \bmod p^k}^* \bar{\chi}(d) e(n\bar{m}d/p^k)$. Since χ is primitive and $(n\bar{m}, p) = 1$, $G_k = 0$ for $k \geq 2$, and $G_1 = \chi(n\bar{m})\tau(\bar{\chi}) = \chi(n)\bar{\chi}(m)\tau(\bar{\chi})$. Using $c_m(n) = \sum_{e|(m, n)} e\mu(m/e)$,

$$D(n, s) = \frac{\tau(\bar{\chi})}{p^{2s} L(2s, \bar{\chi})} \chi(n) \sum_{d|n} \bar{\chi}(d) d^{1-2s}.$$

The coefficients of a Hecke eigenform, normalised at 1, are its eigenvalues for the T_n of Section 1. With $n = ad$, $\chi(n)\bar{\chi}(d) = \chi(a)$ and $n^{s-\frac{1}{2}} d^{1-2s} = (a/d)^{s-\frac{1}{2}}$, we get $\lambda_\infty(n, s) = n^{s-\frac{1}{2}} D(n, s) / D(1, s) = \sum_{ad=n} \chi(a) (a/d)^{s-\frac{1}{2}}$.

Pairing. By the Maass–Selberg relations, the truncated norm of $E_{\mathfrak{a}}(\cdot, \frac{1}{2} + ir)$ is $2 \log Y - \operatorname{Re} \sum_{\mathfrak{c}} \varphi'_{\mathfrak{ac}} \overline{\varphi_{\mathfrak{ac}}}$ plus oscillatory terms. These vanish here because Φ has zero diagonal. The sum runs over row $\mathfrak{a}$ of Φ , since the constant term of $E_{\mathfrak{a}}$ at the cusp $\mathfrak{c}$ is $\delta_{\mathfrak{ac}} y^s + \varphi_{\mathfrak{ac}} y^{1-s}$. Row ∞ has only $\varphi_{\infty 0}$, and $|\varphi_{\infty 0}| = 1$ on the line, so E_∞ pairs with $\omega_\infty = \frac{d}{ds} \log \varphi_{\infty 0}$, which is real on the line.

By Proposition 3, ω_∞ has $\bar{\chi}$ in its L -functions, so $\omega_\infty(r) = \omega_A(-r)$, where ω_A is computed from $\varphi_{0\infty}$ exactly as in Section 3. Also $\lambda_\infty(n, \frac{1}{2} + ir) = \lambda(n, -r)$. Hence E_∞ contributes $\frac{1}{4\pi} \int h(r) \lambda(n, -r) \omega_A(-r) dr = \frac{1}{4\pi} \int h(r) \lambda(n, r) \omega_A(r) dr$, since h is even. E_0 is its reflection under $s \mapsto 1 - s$ and contributes the same amount, which gives the stated $\frac{1}{2\pi}$. $\square$

Check (derive/check_S3.py): $\lambda_\infty(n, s)$ is computed from the Kloosterman-type sums at $s = 2.3$, with $c \leq 600p$, for $n \leq 12$ coprime to p .

- It agrees with $\sum_{ad=n} \chi(a) (a/d)^{s-1/2}$ to 3.4×10^{-12} at $p = 11$ and 5.4×10^{-12} at $p = 7$.
- The other three candidates $(\chi(d), \bar{\chi}(a), \bar{\chi}(d))$ are off by order 1.
- λ_∞ is multiplicative to 2×10^{-10} .

This is the assignment previously selected numerically, now derived.

C The local factor at p for general n (input (S4))

For a nonsplit class with trace t , determinant n , $(n, p) = 1$, and $D = t^2 - 4n = dl^2$ not a square, the geometric side is (archimedean orbital integral) $\times W(t, n)$. The archimedean factor depends only on $t/\sqrt{n}$, as in Section 2. The arithmetic weight is

$$W(t, n) = \sum_{\Gamma_0(p)\text{-classes } [\gamma]} \frac{\chi(a_\gamma)}{|\Gamma_\gamma|} \quad (D < 0),$$

with $|\Gamma_\gamma|$ the order of the stabiliser in PSL_2 . For $D > 0$ the centraliser Γ_γ of γ in $\Gamma_0(p)/\{\pm 1\}$ is infinite cyclic, and the weight is

$$W^+(t, n) = \sum_{\Gamma_0(p)\text{-classes } [\gamma]} \chi(a_\gamma) \ell(\Gamma_\gamma) \quad (D > 0),$$

with $\ell(\Gamma_\gamma)$ the length of the primitive closed geodesic, i.e. the translation length of a generator of Γ_γ .

Proposition 5 ([T]). Write $h_w(\Delta) = h(\Delta)/(\frac{1}{2}w(\Delta))$ for $\Delta < 0$. Then

$$W(t, n) = \sum_{f|l} h_w(D/f^2) \mu(t, f, n), \quad \mu(t, f, n) = \begin{cases} \sum_{x \bmod p, x^2 - tx + n \equiv 0} \chi(x) & p \nmid f, \\ (p+1) \chi(t/2) & p \mid f, \end{cases}$$

and $W(t, n) = H_{t,n} \sum_{f|l} h_w(D/f^2)$ with $H_{t,n}$ as in Theorem 1. Explicitly:

- if $p \nmid D$: $H = \chi(\frac{t+\sqrt{D}}{2}) + \chi(\frac{t-\sqrt{D}}{2})$ when $(\frac{d}{p}) = 1$, and $H = 0$ otherwise;
- if $p \mid D$: $H = \chi(\frac{t}{2})(2 + \frac{k-1}{T})$, where $k = (\frac{d}{p})$, $v = \text{ord}_p l$ and $T = 1 + (p-k)\frac{p^v-1}{p-1}$.

For $D > 0$, write $h^+(\Delta)$ for the number of $\text{SL}_2(\mathbb{Z})$ -classes of primitive forms of discriminant Δ and $\varepsilon_\Delta^+ > 1$ for the generator of the norm-one units of the order of discriminant Δ , $R^+(\Delta) = \log \varepsilon_\Delta^+$. Then, with the same μ and the same $H_{t,n}$,

$$\begin{aligned} W^+(t, n) &= \sum_{f|l} 2h^+(D/f^2) R^+(D/f^2) \mu(t, f, n) \\ &= H_{t,n} \sum_{f|l} 2h^+(D/f^2) R^+(D/f^2) = 2\sqrt{D} H_{t,n} L(1, \psi_D). \end{aligned}$$

Proof. Forms. $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto Q_\gamma = [c, d-a, -b]$ is a bijection from matrices of trace t and determinant n onto forms of discriminant D , with $a = (t-B)/2$. Conjugation by Γ corresponds to equivalence of forms. Every Q_γ is $f \cdot Q'$ with Q' primitive of discriminant D/f^2 , and the condition $p \mid c$ becomes $p \mid fA'$.

Orbits. Each $\text{SL}_2(\mathbb{Z})$ -class of Q' splits into $\Gamma_0(p)$ -classes. These are indexed by the cosets $\sigma \in \Gamma_0(p) \backslash \text{SL}_2(\mathbb{Z}) \cong \mathbb{P}^1(\mathbb{F}_p)$ with $p \mid A(Q' \circ \sigma)$. Precisely: write $Q = fQ' \circ \sigma$; then $Q \circ \gamma = fQ' \circ (\sigma\gamma)$, so the $\Gamma_0(p)$ -classes are the orbits of $S = \text{Aut}^+(Q')$ acting by left multiplication on $\text{SL}_2(\mathbb{Z})/\Gamma_0(p) \cong \mathbb{P}^1(\mathbb{F}_p)$ (first column of σ), and the condition $p \mid fA(Q' \circ \sigma)$ is $fQ'(\alpha, \gamma) \equiv 0 \pmod{p}$. The stabiliser of Q in $\Gamma_0(p)$ is $\sigma^{-1}S\sigma \cap \Gamma_0(p)$, of index $|S\sigma\Gamma_0(p)/\Gamma_0(p)|$ (the orbit length) in $\sigma^{-1}S\sigma$. Since $\gamma \in \Gamma_0(p)$ is upper triangular mod p , $a_\gamma \bmod p$ is a class invariant, so $\chi(a)$ is constant on each orbit.

- $D < 0$: $S/\{\pm 1\}$ is finite, so $1/|\Gamma_\gamma| = |\text{orbit}|/|S/\{\pm 1\}|$, and summing over orbits gives $\frac{1}{|\text{Aut } Q'|} \sum_{\text{cosets}} \chi(a)$.
- $D > 0$: $S/\{\pm 1\}$ is generated by the automorph M of Q' with eigenvalue ε_{D/f^2}^+ [2, Ch. 3], of translation length $2R^+(D/f^2)$. A subgroup of index m in this infinite cyclic group has primitive length $2mR^+$. Hence $\ell(\Gamma_\gamma) = |\text{orbit}| \cdot 2R^+$, and summing over orbits gives $2R^+(D/f^2) \sum_{\text{cosets}} \chi(a)$.

In both cases the weight of a class of Q' is a fixed factor times the sum of $\chi(a)$ over the qualifying cosets, and that sum depends only on $Q' \bmod p$.

Counting. If $p \mid f$, all $p+1$ cosets qualify. Each has $a \equiv t/2 \pmod{p}$, and then $p^2 \mid D$. If $p \nmid f$, the qualifying cosets are the projective roots of $Q' \bmod p$. There are $1 + (\frac{D/f^2}{p})$ of them, and for each one $B^2 \equiv D/f^2 \pmod{p}$. Hence $a = (t-fB)/2$ is a root of $x^2 - tx + n \bmod p$, and distinct roots of Q' give distinct roots. Summing $\chi(a)$ gives $\mu(t, f, n)$.

Local factor. If $p \nmid D$, then $p \nmid l$, μ does not depend on f , and $H = \mu$. If $p \mid D$, both roots equal $t/2$. Write $\sum_{f|l} h_w(D/f^2) = h_w(d) \prod_{q \mid l} c_q$. The p -factor is $c_p = \sum_{j=0}^v c_{p,j}$, with $c_{p,0} = 1$

and $c_{p,j} = p^{j-1}(p-k)$ for $j \geq 1$; here p^j is the p -part of the conductor l/f , so $p \mid f$ exactly when $j \leq v-1$. Hence $H/\chi(\frac{t}{2}) = 1 + p \frac{\sum_{j < v} c_{p,j}}{T}$, with $T = c_p = 1 + (p-k) \frac{p^{v-1}-1}{p-1}$. Moreover $p \sum_{j < v} c_{p,j} = p + p(p-k) \frac{p^{v-1}-1}{p-1} = T + k - 1$, which gives $2 + \frac{k-1}{T}$.

For $D > 0$ the same computation applies to $\sum_f h^+ R^+(D/f^2)$, because $h^+(\Delta)R^+(\Delta) = 2h(\Delta)R(\Delta)$ for every order (if the fundamental unit has norm -1 then $h^+ = h$ and $\varepsilon^+ = \varepsilon^2$, otherwise $h^+ = 2h$ and $\varepsilon^+ = \varepsilon$), and $h(dq^{2j})R(dq^{2j}) = h(d)R(d)q^{j-1}(q-k)$ for $j \geq 1$, $k = (\frac{d}{q})$ (the unit index cancels in the product). So c_p , T and H are unchanged.

Finally, by the class number formula,

$$\sum_{f|l} h_w(D/f^2) = \frac{\sqrt{|D|}}{\pi} L(1, \psi_D) \quad (D < 0), \quad \sum_{f|l} h^+ R^+(D/f^2) = \sqrt{D} L(1, \psi_D) \quad (D > 0),$$

with the same local factors as in Section 1. The archimedean factors then give the two terms of $\mathcal{H}_n$; for $D > 0$ one has $\sqrt{D/n} = 2 \sinh(u/2)$, where $u = 2 \log((|t| + \sqrt{D})/(2\sqrt{n}))$. That normalisation agrees with [1, Thm. 1.7] at $n = \pm 1$, where $H_{t,\pm 1}$ is their factor. $\square$

Check (derive/check_S4.py): for $D < 0$, $p \in \{7, 11\}$, $n < 160$ and all $|t| < 2\sqrt{n}$, three numbers are compared.

- The brute-force count, over reduced forms and the $p+1$ cosets.
- The Cohen–Oesterlé weight $\sum_f h_w \mu$.
- $H_{t,n} \sum_f h_w$.

They agree to 2.6×10^{-14} in 9478 cases, of which 998 have $p \parallel D$ and 104 have $p^2 \mid D$.

Check, $D > 0$ (derive/check_S4_pos.py): $p \in \{7, 11\}$, $n < 160$, $2\sqrt{n} < |t| \leq 2\sqrt{n} + 40$, D not a square. The brute-force side uses no class-group software. The classes of each Q' are the cycles of reduced indefinite forms under Gauss's ρ . M is the product of the ρ -matrices around the cycle. The $\Gamma_0(p)$ -classes are found by iterating M on $\mathbb{P}^1(\mathbb{F}_p)$, with weight $\chi(a) \cdot |\text{orbit}| \cdot 2 \operatorname{arccosh}(|\operatorname{tr} M|/2)$, and $\chi(a)$ is checked to be constant on every orbit. This is compared with $\sum_f 4h(D/f^2)R(D/f^2)\mu$ (PARI `quadclassunit`) and with $2\sqrt{D} H_{t,n} L(1, \psi_D)$ as implemented. They agree to relative error 4.6×10^{-15} in 21590 cases (10346 with nonzero weight; 2316 with $p \parallel D$, 198 with $p^2 \mid D$; 1645 discriminants, 5774 classes). In every class $|\operatorname{tr} M| = \operatorname{tr} \varepsilon^+$ from PARI `quadunit`, and $h^+ R^+$ from the cycles equals $2hR$ to 9×10^{-15} . Control, in the same script: dropping the orbit length (weight $\chi(a) \cdot 2R^+$ per orbit) fails in 194 cases, with relative error up to 0.42, so the check detects the stabiliser index. It bites only in those cases because M preserves Q' , hence permutes its at most two roots mod p ; orbits of length > 1 among qualifying cosets need a swapped root pair or $p \mid f$. In both scripts the Cohen side uses μ exactly as stated in Proposition 5, including $(p+1)\chi(t/2)$ for $p \mid f$. The scripts are a check of the proposition, not a second proof of it.

References

- [1] A. R. Booker, M. Lee, A. Strömbergsson, *Twist-minimal trace formulas and the Selberg eigenvalue conjecture*, J. London Math. Soc. (2020), arXiv:1803.06016.
- [2] D. A. Buell, *Binary Quadratic Forms*, Springer, 1989.
- [3] A. Seymour-Howell, *Rigorous computation of Maass cusp forms of squarefree level*, arXiv:2201.08760.